SUPPLEMENTARY MATERIAL

VIII. NOTATIONS AND MATHEMATICAL DEFINITIONS

In this section, we introduce some symbol definitions and the preliminary knowledge, and supplement the specific definitions of related concepts mentioned in Section II.

A. Definitions of tensor norm

We first introduce the definitions of tensor norms. Given a tensor $\mathcal{M} \in \mathbb{R}^{\prod_{i=1}^{N} I_i}$, let $m_{i_1 i_2 \ldots . i_N}$ denote the $(i_1, i_2, \ldots . i_N)$-th component of $\mathcal{M}$, then the Frobenius norm of $\mathcal{M}$ is defined as

$$\|\mathcal{M}\|_F := \sqrt{\sum_{i_N=1}^{I_N} \ldots . \sum_{i_1=1}^{I_1} m_{i_1 i_2 \ldots . i_N}^2}.$$

From [1], the ℓ_1-norm of $\mathcal{M}$ is defined as

$$\|\mathcal{M}\|_1 := \sum_{i_N=1}^{I_N} \ldots . \sum_{i_1=1}^{I_1} |m_{i_1 i_2 \ldots . i_N}|.$$

B. Mathematical definitions for nonconvex analysis

We first define $\mathbb{Z}$ as the set of all integer numbers, $\lceil a \rceil$ $(a \in \mathbb{R})$ as $\lceil a \rceil := \min\{n \in \mathbb{Z} \mid n \geq a\}$, and $\langle \cdot, \cdot \rangle$ as the inner product.

Next, we recall several basic concepts for nonconvex analysis [2], [3].

Definition 1. *Let $\boldsymbol{S}$ be a set, the derived set $\boldsymbol{S}'$ is defined as the set of overall cluster points of $\boldsymbol{S}$.*

Proposition 1. *A bounded closed set in finite-dimensional space is a compact set.*

Definition 2. *A function $g : \mathbb{R}^n \to (-\infty, +\infty]$ is said to be proper if $\operatorname{dom} g \neq \emptyset$, where $\operatorname{dom} g = \{x \in \mathbb{R} : \ g(x) < \infty\}$.*

Definition 3. *If $\forall x_k \in \operatorname{dom} f$, we have*

$$\lim_{k\to\infty} x_k = x, \ f(x) \leq \liminf_{k\to\infty} f(x_k) \ ,$$

then f is said to be lower semicontinuous on $\operatorname{dom} f$.

Definition 4. *Let f be a proper lower semicontinuous function, the Fréchet subdifferential of f at x, written $\hat{\partial} f(x)$, is the set of all vectors u which satisfy*

$$\lim_{y\neq x, y\to x} \cdot \frac{f(y)-f(x)-\langle u, \ y-x\rangle}{\|y-x\|} \geq 0,$$

when $x \notin \operatorname{dom} f$, then set $\hat{\partial} f(x) = \emptyset$.

Proposition 2. *If f is a proper lower semicontinuous function and has a local minimum at x^*, then $0 \in \partial f(x^*)$.*

Proposition 3. *Let f be a proper lower semicontinuous function, and g be a continuously differential function, then we have $\forall x \in \operatorname{dom} f$, $\partial(f+g)(x) = \partial f(x) + \nabla g(x)$.*

Proposition 4. *Set f: $R^n \to R$, if $\|\nabla f(x) - \nabla f(y)\| \leq L\|x-y\|$, then $f(y) \leq f(x) + \langle \nabla f(x), y-x\rangle + \frac{L}{2}\|y-x\|^2$, where L is the Lipschitz constant of ∇f.*

Definition 5. *Desingularization function: we denote the desingularization function $\phi : [0, \eta) \to \mathbb{R}_+$ which satisfies these conditions: (i) $\phi(0) = 0$. (ii) ϕ is continuously differentiable on $[0, \eta)$, and ϕ is a concave function from in $(0, \eta)$. (iii) $\phi(s)' > 0$ $(\forall s \in (0, \eta))$.*

Definition 6. *Let f be a proper lower semicontinuous function, f is said to have the KŁ property on $\bar{u} \in dom(\partial f)$, if there exists $\eta \in (0, +\infty]$, and U is the neighborhood of $\bar{u}$, any u in U that satisfies the condition $f(\bar{u}) < f(u) < f(\bar{u}) + \eta$ has the following inequalities established.*

$$\phi'(f(u) - f(\bar{u}))dist(0, \partial f(u)) > 1,$$

where ϕ satisfy (Definition 5). If f has the KŁ property at each point of $\operatorname{dom} \partial f$, then f is a KŁ function.

Proposition 5. *(Uniformized KŁ property) Let Ω be a compact set and f: $\mathbb{R}^n \to \mathbb{R} \cup \{\infty\}$ be a proper and lower semicontinuous function. Assume that f is constant on Ω and is a KŁ function. Then, there exist $\epsilon > 0$, $\eta > 0$ such that $\forall x \in \Omega$ and all x in the intersection $\{x \in \mathbb{R}^n : dist(x, \Omega) < \epsilon\} \cap \{f(\bar{x}) < f(x) < f(\bar{x}) + \eta\}$ one has*

$$\phi'(f(x) - f(\bar{x}))dist(0, \partial f(x)) > 1.$$

IX. CONVERGENCE ANALYSIS

In this section, we prove all the theorems in Section V.

A. proof of Theorem 1

Proof. We first define $vec(\mathcal{X})$ as the vectorization of $\mathcal{X}$ gives a vector, which is obtained by stacking all mode-1 fibers of $\mathcal{X}$.

Next, from Eq. (5), we know that

$$\begin{aligned} &vec(\nabla_{\mathcal{G}} H(\bar{\mathcal{G}}, A_{(N)}) - \nabla_{\mathcal{G}} H(\mathcal{G}, A_{(N)})) = \\ &vec((\bar{\mathcal{G}} - \mathcal{G})(\otimes_{i=N}^{1}((A_i)^T A_i))). \end{aligned}$$

Since $\|\mathcal{G}\|_F = \|\mathrm{vec}(\mathcal{G})\|_F$, thus we infer

$$\begin{aligned} &\|\nabla_{\mathcal{G}} H(\bar{\mathcal{G}}, A_{(N)}) - \nabla_{\mathcal{G}} H(\mathcal{G}, A_{(N)})\|_F \\ &= \|vec((\bar{\mathcal{G}} - \mathcal{G})(\otimes_{i=N}^{1}((A_i)^T A_i)))\|_F \\ &\leq \| \otimes_{i=N}^{1} ((A_i)^T A_i)\|_F \|vec(\bar{\mathcal{G}} - \mathcal{G})\|_F \\ &= L_{\nabla_{\mathcal{G}} H} \|\bar{\mathcal{G}} - \mathcal{G}\|_F, \end{aligned}$$

and from Eq. (5), we also have

$$\begin{aligned} &\|\nabla_{A_n} H(\mathcal{G}, \{A_j\}_{j=1}^{n-1}, \bar{A}, \{A_j\}_{j=n+1}^{N}) - \nabla_{A_n} H(\mathcal{G}, A_{(N)})\|_F \\ &\leq L_{\nabla_{A_n} H} \|\bar{A} - A_n\|_F, \end{aligned}$$

where $B_n = G^{(n)}(\otimes_{i=N, i\neq n}^{1} A_i)^T$ and

$$L_{\nabla_{A_n} H} = \begin{cases} \|B_n (B_n)^T\|_F + \| \sum_{q=1}^{Q} \alpha_q L_q\|_F, & \text{if } n = N, \\ \|B_n (B_n)^T\|_F, & \text{else.} \end{cases}$$

Thus, Theorem 1 is true. □

B. proof of Theorem 2

Proof. (i) If step 9 of Algorithm 1 is not true, then the Theorem 2(i) is true, if step 9 of Algorithm 1 is true, since Theorem 1 and Proposition 4, we know that

$$\begin{aligned} H(\mathcal{G}^{k+1}, A_{(N)}^k) \leq &\langle \nabla_{\mathcal{G}} H(\mathcal{G}^k, A_{(N)}^k), \mathcal{G}^{k+1} - \mathcal{G}^k\rangle \\ &+ H(\mathcal{G}^k, A_{(N)}^k) \\ &+ \frac{L_{\nabla_{\mathcal{G}^k} H}}{2} \|\mathcal{G}^{k+1} - \mathcal{G}^k\|_F^2. \end{aligned} \tag{7}$$

From Eq. (6), we obtain

$$\begin{aligned} F_G(\mathcal{G}^k) \geq &F_G(\mathcal{G}^{k+1}) + \frac{1}{2\sigma_G^k} \|\mathcal{G}^{k+1} - \mathcal{G}^k\|_F^2 \\ &+ \langle \nabla_{\mathcal{G}} H(\mathcal{G}^k, A_{(N)}^k), \mathcal{G}^{k+1} - \mathcal{G}^k\rangle. \end{aligned} \tag{8}$$

Then sum of the Eq. (7) and Eq. (8), we have

$$\begin{aligned} H(\mathcal{G}^{k+1}, A_{(N)}^k) \leq &H(\mathcal{G}^k, A_{(N)}^k) + F_G(\mathcal{G}^k) - F_G(\mathcal{G}^{k+1}) \\ &- (\frac{1}{2\sigma_G^k} - \frac{L_{\nabla_{\mathcal{G}^k} H}}{2}) \|\mathcal{G}^{k+1} - \mathcal{G}^k\|_F^2. \end{aligned} \tag{9}$$

Similarly, since Theorem 1 and Proposition 4, we know that

$$H(\mathcal{G}^{k+1}, A_1^{k+1}, \{A_i^k\}_{i=2}^N) \leq \langle \nabla_{A_1} H(\mathcal{G}^{k+1}, A_{(N)}^k), A_1^{k+1} - A_1^k \rangle + H(\mathcal{G}^{k+1}, A_{(N)}^k) + \frac{L_{\nabla_{A_1^k} H}}{2} \|A_1^{k+1} - A_1^k\|_F^2. \quad (10)$$

From Eq. (6), we can also obtain

$$F_1(A_1^k) \geq F_1(A_1^{k+1}) + \frac{1}{2\sigma_1^k} \|A_1^{k+1} - A_1^k\|_F^2 + \langle \nabla_{A_1} H(\mathcal{G}^{k+1}, A_{(N)}^k), A_1^{k+1} - A_1^k \rangle. \quad (11)$$

Then sum of the Eq. (10), Eq. (11) and Eq. (9), we have

$$\begin{aligned} H(\mathcal{G}^{k+1}, A_1^{k+1}, \{A_i^k\}_{i=2}^N) \leq\ & H(\mathcal{G}^k, A_{(N)}^k) + F_1(A_1^k) + F_G(\mathcal{G}^k) \\ & - F_1(A_1^{k+1}) - F_G(\mathcal{G}^{k+1}) \\ & - \rho \|A_1^{k+1} - A_1^k\|_F^2 \\ & - \rho \|\mathcal{G}^{k+1} - \mathcal{G}^k\|_F^2, \end{aligned} \quad (12)$$

where $\rho = \min(\frac{1}{2\sigma_G^k} - \frac{L_{\nabla_{\mathcal{G}^k} H}}{2}, \frac{1}{2\sigma_1^k} - \frac{L_{\nabla_{A_1^k} H}}{2})$. Assuming Theorem 2(i) holds when $n = N-1$, i.e.,

$$\begin{aligned} H(\mathcal{G}^{k+1}, A_{(N-1)}^{k+1}, A_N^k) \leq\ & H(\mathcal{G}^k, A_{(N)}^k) + F_G(\mathcal{G}^k) + \sum_{i=1}^{N-1} F_i(A_i^k) \\ & - \sum_{i=1}^{N-1} F_i(A_i^{k+1}) - F_G(\mathcal{G}^{k+1}) \\ & - \rho \|A_{(N-1)}^{k+1} - A_{(N-1)}^k\|_F^2 \\ & - \rho \|\mathcal{G}^{k+1} - \mathcal{G}^k\|_F^2, \end{aligned} \quad (13)$$

where $\rho = \min(\frac{1}{2\sigma_G^k} - \frac{L_{\nabla_{\mathcal{G}^k} H}}{2}, \{\frac{1}{2\sigma_i^k} - \frac{L_{\nabla_{A_i^k} H}}{2}\}_{i=1}^{N-1})$. Similarly, from Eq. (6), we infer

$$F_N(A_N^k) \geq F_N(A_N^{k+1}) + \frac{1}{2\sigma_N^k} \|A_N^{k+1} - A_N^k\|_F^2 + \langle \nabla_{A_N} H(\mathcal{G}^{k+1}, A_{(N-1)}^{k+1}, A_N^k), A_N^{k+1} - A_N^k \rangle. \quad (14)$$

From Theorem 1 and Proposition 4, we also have

$$\begin{aligned} H(\mathcal{G}^{k+1}, A_{(N)}^{k+1}) \leq\ & H(\mathcal{G}^{k+1}, A_{(N-1)}^{k+1}, A_N^k) \\ & + \langle \nabla_{A_N} H(\mathcal{G}^{k+1}, A_{(N-1)}^{k+1}, A_N^k), A_N^{k+1} - A_N^k \rangle \\ & + \frac{L_{\nabla_{A_N^k} H}}{2} \|A_N^{k+1} - A_N^k\|_F^2. \end{aligned} \quad (15)$$

Then sum of the Eq. (13), Eq. (14) and Eq. (15), we have

$$J(\mathcal{G}^{k+1}, A_{(N)}^{k+1}) \leq J(\mathcal{G}^k, A_{(N)}^k) - \rho \|z^{k+1} - c^k\|_F^2, \quad (16)$$

where $\rho = \min(\frac{1}{2\sigma_G^k} - \frac{L_{\nabla_{\mathcal{G}^k} H}}{2}, \{\frac{1}{2\sigma_i^k} - \frac{L_{\nabla_{A_i^k} H}}{2}\}_{i=1}^N)$. This shows the Theorem 2(i) is true.

(ii) We know that $\frac{\alpha_q}{2} \operatorname{Tr}((A_N)^T L_q A_N)$ in Eq. (4) is nonnegative. Hence, Eq. (4) is nonnegative. From Eq. (16), we have

$$\rho \|z^{k+1} - c^k\|_F^2 \leq J(\mathcal{G}^k, A_{(N)}^k) - J(\mathcal{G}^{k+1}, A_{(N)}^{k+1}).$$

Sum of both side, since Eq. (4) is nonnegative, thus we have

$$\begin{aligned} \rho \sum_{k=1}^{\infty} \|z^{k+1} - c^k\|_F^2 \leq\ & \sum_{k=1}^{\infty} (J(\mathcal{G}^k, A_{(N)}^k) - J(\mathcal{G}^{k+1}, A_{(N)}^{k+1})) \\ =\ & J(\mathcal{G}^1, A_{(N)}^1) - \inf J \\ <\ & \infty. \end{aligned}$$

It follows that

$$\lim_{k \to \infty} \|z^{k+1} - c^k\|_F^2 = 0.$$

□

C. Lemma: subgradient lower bound property of Algorithm 1

Before continuing with the proof of the theorems of Section V, we prove the following lemma.

Lemma 1. *If we define*

$$\begin{aligned} f_G^{k+1} &= \nabla_{\mathcal{G}} H(\mathcal{G}^{k+1}, A_{(N)}^k) + \frac{1}{\sigma_G^k}(\mathcal{Y}^k - \mathcal{G}^{k+1}) \\ &\quad - \nabla_{\mathcal{G}} H(\mathcal{Y}^k, A_{(N)}^k), \\ f_i^{k+1} &= \nabla_{A_i} H(\mathcal{G}^{k+1}, \{A_j^{k+1}\}_{j=1}^i, \{A_j^k\}_{j=i+1}^N) \\ &\quad - \nabla_{A_i} H(\mathcal{G}^{k+1}, \{A_j^{k+1}\}_{j=1}^{i-1}, Y_i^k, \{A_j^k\}_{j=i+1}^N), \\ &\quad + \frac{1}{\sigma_i^k}(Y_i^k - A_i^{k+1}). \end{aligned} \quad (17)$$

Then we have

$$\|\{f_i^{k+1}\}_{i=1}^N\|_F + \|f_G^{k+1}\|_F \leq \rho_b \|z^{k+1} - c^k\|_F,$$

where ρ_b is a positive constant.

Proof. By Proposition 2 and Proposition 3, Eq. (6) follows that

$$0 \in \partial_{\mathcal{G}} F_G(\mathcal{G}^{k+1}) - \frac{1}{\sigma_G^k}(\mathcal{Y}^k - \mathcal{G}^{k+1}) + \nabla_{\mathcal{G}} H(\mathcal{Y}^k, A_{(N)}^k). \quad (18)$$

Thus we infer

$$\begin{aligned} f_G^{k+1} &= \nabla_{\mathcal{G}} H(\mathcal{G}^{k+1}, A_{(N)}^k) - \nabla_{\mathcal{G}} H(\mathcal{Y}^k, A_{(N)}^k) \\ &\quad + \frac{1}{\sigma_G^k}(\mathcal{Y}^k - \mathcal{G}^{k+1}) \\ &\in \nabla_{\mathcal{G}} H(\mathcal{G}^{k+1}, A_{(N)}^k) + \partial_{\mathcal{G}}(F_G(\mathcal{G}^{k+1}) + \sum_{i=1}^N F_i(A_i^k)) \\ &= \partial_{\mathcal{G}} J(\mathcal{G}^{k+1}, A_{(N)}^k). \end{aligned} \quad (19)$$

Similarly, by Proposition 2 and Proposition 3, Eq. (6) follows that

$$\begin{aligned} 0 &\in \partial_{A_i} F_i(A_i^{k+1}) - \frac{1}{\sigma_i^k}(Y_i^k - A_i^{k+1}) \\ &\quad + \nabla_{A_i} H(\mathcal{G}^{k+1}, \{A_j^{k+1}\}_{j=1}^{i-1}, Y_i^k, \{A_j^k\}_{j=i+1}^N). \end{aligned} \quad (20)$$

Thus we also have

$$\begin{aligned} f_i^{k+1} &= \nabla_{A_i} H(\mathcal{G}^{k+1}, \{A_j^{k+1}\}_{j=1}^i, \{A_j^k\}_{j=i+1}^N) \\ &\quad - \frac{1}{\sigma_i^k}(A_i^{k+1} - Y_i^k) \\ &\quad - \nabla_{A_i} H(\mathcal{G}^{k+1}, \{A_j^{k+1}\}_{j=1}^{i-1}, Y_i^k, \{A_j^k\}_{j=i+1}^N) \\ &\in \nabla_{A_i} H(\mathcal{G}^{k+1}, \{A_j^{k+1}\}_{j=1}^{i-1}, A_i^{k+1}, \{A_j^k\}_{j=i+1}^N) \\ &\quad + \partial_{A_i}(F_G(\mathcal{G}^{k+1}) + \sum_{j=1}^i F_j(A_j^{k+1}) + \sum_{j=i+1}^N F_j(A_j^k)) \\ &= \partial_{A_i} J(\mathcal{G}^{k+1}, \{A_j^{k+1}\}_{j=1}^{i-1}, A_i^{k+1}, \{A_j^k\}_{j=i+1}^N). \end{aligned} \quad (21)$$

Since Theorem 1, we have

$$\begin{aligned}\|f_G^{k+1}\|_F &= \|\nabla_{\mathcal{G}} H(\mathcal{G}^{k+1}, A_{(N)}^k) + \frac{1}{\sigma_G^k}(\mathcal{Y}^k - \mathcal{G}^{k+1}) \\ &\quad - \nabla_{\mathcal{G}} H(\mathcal{Y}^k, A_{(N)}^k)\|_F, \\ &\le \|\nabla_{\mathcal{G}} H(\mathcal{G}^{k+1}, A_{(N)}^k) - \nabla_{\mathcal{G}} H(\mathcal{Y}^k, A_{(N)}^k)\|_F \\ &\quad + \|\frac{1}{\sigma_G^k}(\mathcal{Y}^k - \mathcal{G}^{k+1})\|_F \\ &\le (L_{\nabla_{\mathcal{G}^k} H} + \frac{1}{\sigma_G^k})\|\mathcal{G}^{k+1} - \mathcal{Y}^k\|_F. \end{aligned} \tag{22}$$

Similarly, we also have

$$\begin{aligned}\|f_i^{k+1}\|_F &= \|\nabla_{A_i} H(\mathcal{G}^{k+1}, \{A_j^{k+1}\}_{j=1}^i, \{A_j^k\}_{j=i+1}^N) \\ &\quad + \frac{1}{\sigma_i^k}(Y_i^k - A_i^{k+1}) \\ &\quad - \nabla_{A_i} H(\mathcal{G}^{k+1}, \{A_j^{k+1}\}_{j=1}^{i-1}, Y_i^k, \{A_j^k\}_{j=i+1}^N)\|_F, \\ &\le \|\nabla_{A_i} H(\mathcal{G}^{k+1}, \{A_j^{k+1}\}_{j=1}^i, \{A_j^k\}_{j=i+1}^N) \\ &\quad - \nabla_{A_i} H(\mathcal{G}^{k+1}, \{A_j^{k+1}\}_{j=1}^{i-1}, Y_i^k, \{A_j^k\}_{j=i+1}^N)\|_F \\ &\quad + \|\frac{1}{\sigma_i^k}(A_i^{k+1} - Y_i^k)\|_F \\ &\le (L_{\nabla_{A_i^k} H} + \frac{1}{\sigma_i^k})\|A_i^{k+1} - Y_i^k\|_F. \end{aligned} \tag{23}$$

Thus we infer

$$\|f_G^{k+1}\|_F + \|\{f_i^{k+1}\}_{i=1}^N\|_F \le \rho_b \|z^{k+1} - c^k\|_F, \tag{24}$$

where $\rho_b = \max(\frac{1}{\sigma_G^k} + L_{\nabla_{\mathcal{G}^k} H}, \{\frac{1}{\sigma_i^k} + L_{\nabla_{A_i^k} H}\}_{i=1}^N)$. □

D. proof of Theorem 3

Proof. (i) Observe that z' can be viewed as an intersection of compact sets

$$z' = \cap_{q\in\mathbb{N}} \overline{\cup_{k\ge q}\{z^k\}}.$$

From proposition 1, z' is compact.

(ii) $\forall \overline{z} \in z'$, there exists a subsequence z^{k_j} such that

$$\lim_{j\to\infty} z^{k_j} = \overline{z}.$$

Let

$$F(z^{k_j}) = \sum_{i=1}^N F_i(A_i^{k_j}) + F_G(\mathcal{G}^{k_j}). \tag{25}$$

Since F_i and F_G are lower semicontinuous [3], from Definition 3 and Eq. (25), we obtain that

$$\lim_{j\to\infty} \inf F(z^{k_j}) \ge F(\overline{z}). \tag{26}$$

Choosing $k = k_j - 1$, from Eq. (6), we infer

$$\begin{aligned}\lim_{j\to\infty} \sup F_G(\mathcal{G}^{k_j}) &\le \lim_{j\to\infty} \sup(F_G(\overline{\mathcal{G}}) + \frac{1}{2\sigma_G^k}\|\overline{\mathcal{G}} - \mathcal{Y}^k\|_F^2 \\ &\quad + \langle \nabla_{\mathcal{G}} H(\mathcal{Y}^k, A_{(N)}^k), \overline{\mathcal{G}} - \mathcal{Y}^k\rangle), \\ \lim_{j\to\infty} \sup F_i(A_i^{k_j}) &\le \lim_{j\to\infty} \sup(F_i(\overline{A_i}) + \frac{1}{2\sigma_i^k}\|\overline{A_i} - Y_i^k\|_F^2 \\ &\quad + \langle \nabla_{A_i} h_i(Y_i^k), \overline{A_i} - Y_i^k\rangle), \end{aligned} \tag{27}$$

where $h_i(Y_i^k) = H(\mathcal{G}^{k+1}, \{A_n^{k+1}\}_{n=1}^{i-1}, Y_i^k, \{A_n^k\}_{n=i+1}^N)$, $\lim_{j\to\infty} \mathcal{G}^{k_j} = \overline{\mathcal{G}}$, $\lim_{j\to\infty} A_i^{k_j} = \overline{A_i}$. Since $\lim_{j\to\infty} z^{k_j} = \overline{z}$ and $\lim_{j\to\infty} \|z^{k_j} - c^{k_j-1}\|_F = 0$ (Theorem 2(ii)), we can get that

$$\begin{aligned}\lim_{j\to\infty} \|\overline{A_i} - Y_i^k\|_F &\le \lim_{j\to\infty} \|\overline{A_i} - A_i^{k_j}\|_F \\ &\quad + \lim_{j\to\infty} \|A_i^{k_j} - Y_i^{k_j-1}\|_F \\ &= 0, \\ \lim_{j\to\infty} \|\overline{\mathcal{G}} - \mathcal{Y}^k\|_F &\le \lim_{j\to\infty} \|\overline{\mathcal{G}} - \mathcal{G}^{k_j}\|_F \\ &\quad + \lim_{j\to\infty} \|\mathcal{G}^{k_j} - \mathcal{Y}^{k_j-1}\|_F \\ &= 0. \end{aligned} \tag{28}$$

From Eq. (27) and Eq. (28), we infer

$$\begin{aligned}\lim_{j\to\infty} \sup F_i(A_i^{k_j}) &\le \lim_{j\to\infty} \sup F_i(\overline{A_i}), \\ \lim_{j\to\infty} \sup F_G(\mathcal{G}^{k_j}) &\le \lim_{j\to\infty} \sup F_G(\overline{\mathcal{G}}), \end{aligned} \tag{29}$$

where $\lim_{j\to\infty} \mathcal{G}^{k_j} = \overline{\mathcal{G}}$, $\lim_{j\to\infty} A_i^{k_j} = \overline{A_i}$. Therefore, from Eq. (25) and Eq. (29), we have

$$\lim_{j\to\infty} \sup F(z^{k_j}) \le \lim_{j\to\infty} \sup F(\overline{z}). \tag{30}$$

From Eq. (25), Eq. (30) and Eq. (26), we infer

$$\lim_{j\to\infty} F(z^{k_j}) = F(\overline{z}). \tag{31}$$

Since Theorem 2, we have

$$\lim_{j\to\infty} H(z^{k_j}) = H(\overline{z}). \tag{32}$$

Thus from Eq. (31) and Eq. (32), we infer

$$\begin{aligned}\lim_{j\to\infty} H(z^{k_j}) + \lim_{j\to\infty} F(z^{k_j}) &= \lim_{j\to\infty} (H(z^{k_j}) + F(z^{k_j})) \\ &= \lim_{j\to\infty} J(z^{k_j}) \\ &= J(\overline{z}). \end{aligned}$$

This means J is constant on z'.

(iii) From Eq. (19) and Eq. (21), we know that

$$\begin{aligned} f_G^{k+1} &\in \partial_{\mathcal{G}} J(\mathcal{G}^{k+1}, A_{(N)}^k), \\ f_i^{k+1} &\in \partial_{A_i} J(\mathcal{G}^{k+1}, \{A_j^{k+1}\}_{j=1}^i, \{A_j^k\}_{j=i+1}^N), \end{aligned} \tag{33}$$

where the definitions of f_i^{k+1} and f_G^{k+1} are the same as in Eq. (17). From Theorem 2(ii) and Lemma 1, we infer

$$\begin{aligned}\lim_{k\to\infty} (\|\{f_i^{k+1}\}_{i=1}^N\|_F + \|f_G^{k+1}\|_F) &\le \lim_{k\to\infty} \rho_b \|z^{k+1} - c^k\|_F \\ &= 0. \end{aligned} \tag{34}$$

Let

$$z_1^k = \mathcal{G}^k, z_{i+1}^k = A_i^k. \tag{35}$$

From Theorem 2(i), we know that

$$\begin{aligned}\lim_{k\to\infty} J(\left\{z_i^{k+1}\right\}_{i=1}^{j-1}, z_j^{k+1}, \left\{z_i^k\right\}_{i=j+1}^{N+1}) &= \lim_{k\to\infty} J(z^k) \\ &= J(z^*). \end{aligned} \tag{36}$$

Therefore, from Eq. (33), Eq. (34), Eq. (35) and Eq. (36), we have

$$\begin{aligned} 0 &\in \partial_{z_j} \lim_{k\to\infty} J(\left\{z_i^{k+1}\right\}_{i=1}^{j-1}, z_j^{k+1}, \left\{z_i^k\right\}_{i=j+1}^{N+1}) \\ &= \partial_{z_j} J(z^*). \end{aligned}$$

This means $0 \in \partial J(z^*)$. □

E. proof of Theorem 4

Proof. From Theorem 1, we know that $\sum_{q=1}^{Q} \frac{\alpha_q}{2} \operatorname{Tr}((A_N)^T L_q A_N)$ is an infinitely differentiable function, and the norm of its derivatives of any order is also continuous, so $\sum_{q=1}^{Q} \frac{\alpha_q}{2} \operatorname{Tr}((A_N)^T L_q A_N)$ is a real analytic function (real analytic functions are all KŁ functions [4]). Since the Frobenius norm, ℓ_p-norm is also all KŁ functions [3], it follows that Eq. (4) is a KŁ function. Therefore, from Definition 6, there exists a concave function ϕ so that

$$\phi^{'}(J(z^k) - J(\overline{z}))dist(0, \partial J(z^k)) \geq 1. \tag{37}$$

From ϕ is the convex function, we have

$$\begin{aligned}\phi(J(z^{k+1}) - J(\overline{z})) \leq \phi(J(z^k) - J(\overline{z})) \\ + \phi^{'}(J(z^k) - J(\overline{z}))(J(z^{k+1}) - J(z^k)).\end{aligned} \tag{38}$$

From Lemma 1 and Theorem 3, we infer

$$dist(0, \partial J(z^k)) \leq \rho_b \|z^k - c^{k-1}\|_F. \tag{39}$$

Since the Eq. (37) and Eq. (39), we infer

$$\begin{aligned}\phi^{'}(J(z^k) - J(\overline{z})) &\geq \frac{1}{dist(0, \partial J(z^k))} \\ &\geq \frac{1}{\rho_b \|z^k - c^{k-1}\|_F}.\end{aligned} \tag{40}$$

Let $J(k) = J(z^k) - J(\overline{z})$, from Eq. (38), Eq. (39), Eq. (40) and Theorem 2(i), we have

$$\begin{aligned}\phi(J(k)) - \phi(J(k+1)) &\geq \phi^{'}(J(k))(J(k) - J(k+1)) \\ &\geq \frac{J(k) - J(k+1)}{\rho_b \|z^k - c^{k-1}\|_F} \\ &\geq \frac{\rho \|z^{k+1} - c^k\|_F^2}{\rho_b \|z^k - c^{k-1}\|_F}.\end{aligned}$$

Define C=$\frac{\rho}{\rho_b}$, C is a constant, so we infer

$$\|z^{k+1} - c^k\|_F^2 \leq C(\phi(J(k)) - \phi(J(k+1)))\|z^k - c^{k-1}\|_F.$$

Using the fact that $2ab \leq a^2 + b^2$

$$2\|z^{k+1} - c^k\|_F \leq C(\phi(J(k)) - \phi(J(k+1))) + \|z^k - c^{k-1}\|_F.$$

Sum both sides

$$\begin{aligned}2\sum_{k=l+1}^{K} \|z^{k+1} - c^k\|_F \leq \sum_{k=l+1}^{K} \|z^k - c^{k-1}\|_F \\ + C(\phi(J(l+1)) - \phi(J(K+1))) \\ = C(\phi(J(l+1)) - \phi(J(K+1))) \\ + \|z^{l+1} - c^l\|_F + \sum_{k=l+1}^{K} \|z^{k+1} - c^k\|_F.\end{aligned} \tag{41}$$

From Eq. (41), we can get that

$$\begin{aligned}\lim_{K\to\infty} \sum_{k=l+1}^{K} \|z^{k+1} - c^k\|_F \leq \|z^{l+1} - c^l\|_F + C\phi(J(l+1)) \\ - \lim_{K\to\infty} C\phi(J(K+1)) \\ < \infty.\end{aligned}$$

Thus we have

$$\sum_{k=l+1}^{\infty} \|z^{k+1} - c^k\|_F < \infty. \tag{42}$$

From Algorithm 1, no matter $c^k = z^k + \beta_k(z^k - z^{k-1})$ or $c^k = z^k$, we always have

$$\|z^{k+1} - z^k\|_F - \beta_k \|z^k - z^{k-1}\|_F \leq \|z^{k+1} - c^k\|_F. \tag{43}$$

From Eq. (42) and Eq. (43), we know that

$$\begin{aligned}\sum_{k=l+1}^{\infty} (\|z^{k+1} - z^k\|_F - \beta_k \|z^k - z^{k-1}\|_F) \leq \sum_{k=l+1}^{\infty} \|z^{k+1} - c^k\|_F \\ < \infty.\end{aligned}$$

Since

$$\begin{aligned}\|z^{k+1} - z^k\|_F - \beta_{max} \|z^k - z^{k-1}\|_F \leq \\ \|z^{k+1} - z^k\|_F - \beta_k \|z^k - z^{k-1}\|_F,\end{aligned}$$

we also have

$$\begin{aligned}&\sum_{k=l+1}^{\infty} (\|z^{k+1} - z^k\|_F - \beta_{max} \|z^k - z^{k-1}\|_F) \\ &\leq \sum_{k=l+1}^{\infty} (\|z^{k+1} - z^k\|_F - \beta_k \|z^k - z^{k-1}\|_F) \\ &< \infty,\end{aligned} \tag{44}$$

and

$$\begin{aligned}&\sum_{k=l+1}^{\infty} \|z^{k+1} - z^k\|_F - \sum_{k=l+1}^{\infty} \beta_{max} \|z^k - z^{k-1}\|_F \\ &= \sum_{k=l+1}^{\infty} (\|z^{k+1} - z^k\|_F - \beta_{max} \|z^{k+1} - z^k\|_F) \\ &+ \beta_{max} \|z^{l+1} - z^l\|_F \\ &= \sum_{k=l+1}^{\infty} (1 - \beta_{max}) \|z^{k+1} - z^k\|_F + \beta_{max} \|z^{l+1} - z^l\|_F.\end{aligned} \tag{45}$$

Let $s^k = \|z^{k+1} - z^k\|_F$, from Eq. (44) and Eq. (45), we have

$$\begin{aligned}\sum_{k=l+1}^{\infty} (1 - \beta_{max}) s^k - \beta_{max} s^l = \sum_{k=l}^{\infty} (s^{k+1} - \beta_{max} s^k) \\ \leq \sum_{k=l+1}^{\infty} \|z^{k+1} - c^k\|_F \\ < \infty.\end{aligned}$$

Thus we infer

$$\sum_{k=l+1}^{\infty} (1 - \beta_{max}) s^k < \infty. \tag{46}$$

From $(1 - \beta_{max})$ is a positive constant and Eq. (46), we infer

$$\sum_{k=l+1}^{\infty} \|z^{k+1} - z^k\|_F = \sum_{k=l+1}^{\infty} s^k < \infty.$$

Thus we have

$$\sum_{k=0}^{\infty} \|z^{k+1} - z^k\|_F = \sum_{k=0}^{\infty} s^k < \infty.$$

Table IV: Average and standard deviation results by all methods on the remaining datasets. The best performance is highlighted in bold.

Method \ Dataset	USPS			ConcreteCrackImages		
	ACC	ARI	NMI	ACC	ARI	NMI
ℓ_1-SNTD	0.627 ± 0.015	0.465 ± 0.022	0.566 ± 0.013	0.788 ± 0.003	0.342 ± 0.006	0.385 ± 0.012
ARTD	0.644 ± 0.017	0.482 ± 0.021	0.581 ± 0.013	0.802 ± 0.010	0.366 ± 0.024	0.404 ± 0.020
GSNTD	0.640 ± 0.022	0.477 ± 0.023	0.586 ± 0.013	0.796 ± 0.020	0.361 ± 0.027	0.416 ± 0.012
ℓ_0-GSNTD	0.668 ± 0.014	0.489 ± 0.013	0.596 ± 0.019	0.815 ± 0.002	0.396 ± 0.005	0.432 ± 0.002
MGNTD	0.565 ± 0.019	0.417 ± 0.025	0.534 ± 0.019	0.804 ± 0.015	0.369 ± 0.036	0.410 ± 0.026
AMGRNTD	0.586 ± 0.033	0.433 ± 0.033	0.541 ± 0.031	0.805 ± 0.003	0.370 ± 0.004	0.429 ± 0.007
SMGNTD	**0.687 ± 0.020**	**0.492 ± 0.020**	**0.599 ± 0.011**	**0.816 ± 0.001**	**0.399 ± 0.004**	**0.447 ± 0.003**

Method \ Dataset	PNW			OrganSMNIST		
	ACC	ARI	NMI	ACC	ARI	NMI
ℓ_1-SNTD	0.464 ± 0.006	0.127 ± 0.009	0.307 ± 0.023	0.290 ± 0.005	0.131 ± 0.006	0.271 ± 0.007
ARTD	0.465 ± 0.017	0.117 ± 0.027	0.350 ± 0.039	0.304 ± 0.007	0.136 ± 0.006	0.277 ± 0.005
GSNTD	0.467 ± 0.013	0.119 ± 0.006	0.351 ± 0.010	0.296 ± 0.006	0.131 ± 0.006	0.272 ± 0.005
ℓ_0-GSNTD	0.454 ± 0.008	0.115 ± 0.006	0.342 ± 0.006	0.307 ± 0.007	0.139 ± 0.006	0.288 ± 0.003
MGNTD	0.451 ± 0.015	0.108 ± 0.017	0.282 ± 0.013	0.273 ± 0.020	0.104 ± 0.018	0.243 ± 0.014
AMGRNTD	0.457 ± 0.012	0.111 ± 0.007	0.288 ± 0.008	0.295 ± 0.013	0.133 ± 0.005	0.286 ± 0.004
SMGNTD	**0.472 ± 0.005**	**0.138 ± 0.009**	**0.355 ± 0.004**	**0.312 ± 0.006**	**0.142 ± 0.003**	**0.293 ± 0.002**

This shows that

$$\begin{aligned}\lim_{K\to\infty}\|z^{K+p}-z^K\|_F &= \lim_{K\to\infty}\|\sum_{k=K+1}^{K+p}(z^{k+1}-z^k)\|_F\\ &\le \lim_{K\to\infty}\sum_{k=K+1}^{K+p}\|z^{k+1}-z^k\|_F\\ &= \lim_{K\to\infty}\sum_{k=K+1}^{\infty}s^k\\ &= 0. \end{aligned} \tag{47}$$

This means the Theorem 4 is true. □

F. proof of Theorem 5

Proof. From Appendix B of [1] and Section 3.5 of [5], we know that the computational cost of $\nabla_{A_i}H(\mathcal{G},A_{(N)})$ is

$$\mathcal{O}(\sum_{j=1}^{N}(\sum_{i=1}^{j}I_i)(\sum_{i=j}^{N}r_i)). \tag{48}$$

Similarly, the computational cost of $\nabla_{\mathcal{G}}H(\mathcal{G},A_{(N)})$ is

$$\mathcal{O}(\sum_{j=1}^{N}(\sum_{i=1}^{j}r_i)(\sum_{i=j}^{N}I_i)). \tag{49}$$

The cost of computing the Lipschitz constant, projection to nonnegative, and tensor unfolding is negligible compared to the cost of computing partial gradients $\nabla_{\mathcal{G}}H(\mathcal{G},A_{(N)})$ and $\nabla_{A_j}H(\mathcal{G},A_{(N)})$. Therefore, from Eq. (48) and Eq. (49), the time complexity of the Algorithm 1 in each iteration is approximately estimated as follows.

$$\mathcal{O}(\sum_{j=1}^{N}(\sum_{i=1}^{j}r_i)(\sum_{i=j}^{N}I_i)+N(\sum_{j=1}^{N}(\sum_{i=1}^{j}I_i)(\sum_{i=j}^{N}r_i))).$$

□

X. Supplementary Experimental Results

In this section, we continue to complete the remaining experiments outlined in Section VI.

A. Supplementary experiment 1: Clustering performance evaluations on the remaining datasets

We continue to evaluate the clustering performance of all methods on the remaining four real-world datasets described in Section VI, and all parameter settings are the same as in Section VI. Table IV shows the clustering performance results (average and standard deviation) for the methods over 10 independent runs with different initial points on the remaining datasets. From Table IV, we can get the same conclusion as Section VI that SMGNTD remarkably outperforms all other methods on all datasets. This once again fully demonstrates the effectiveness of SMGNTD and shows that simultaneously integrating sparse learning and multi-graph learning techniques into NTD can indeed enhance both the feature extraction capability and interpretability.

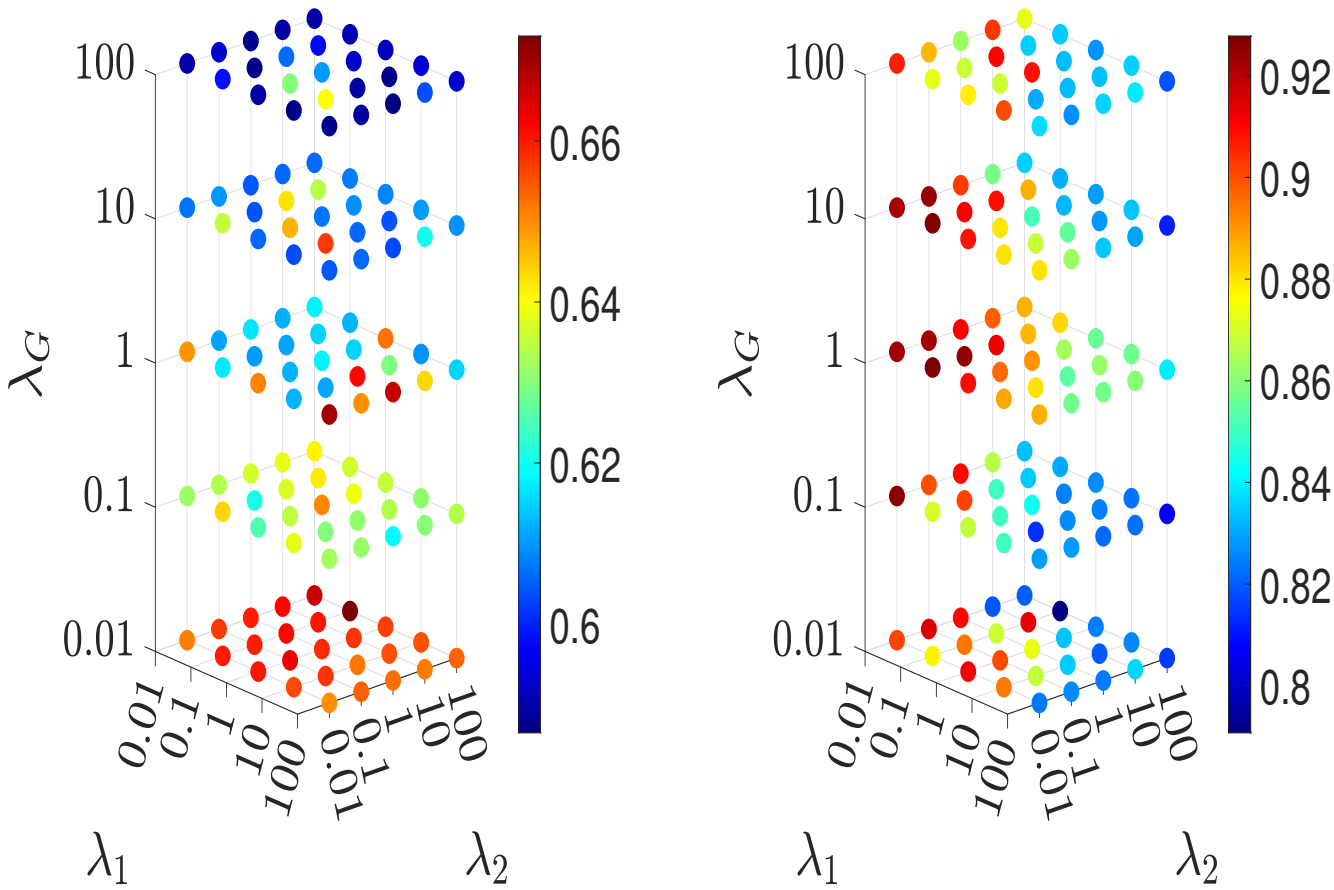
B. Supplementary experiment 2: Parameter sensitivity analysis

As outlined in Section VI, we continue to analyze the impact of both the sparsity level parameters λ_G, λ_i $(\forall i\in[N-1])$ and the Tucker rank r_i $(\forall i\in[N-1])$ in SMGNTD on clustering performance. Specifically, we report the clustering performance under the following two parameter settings on the FashionMNIST and Pixraw10P datasets.

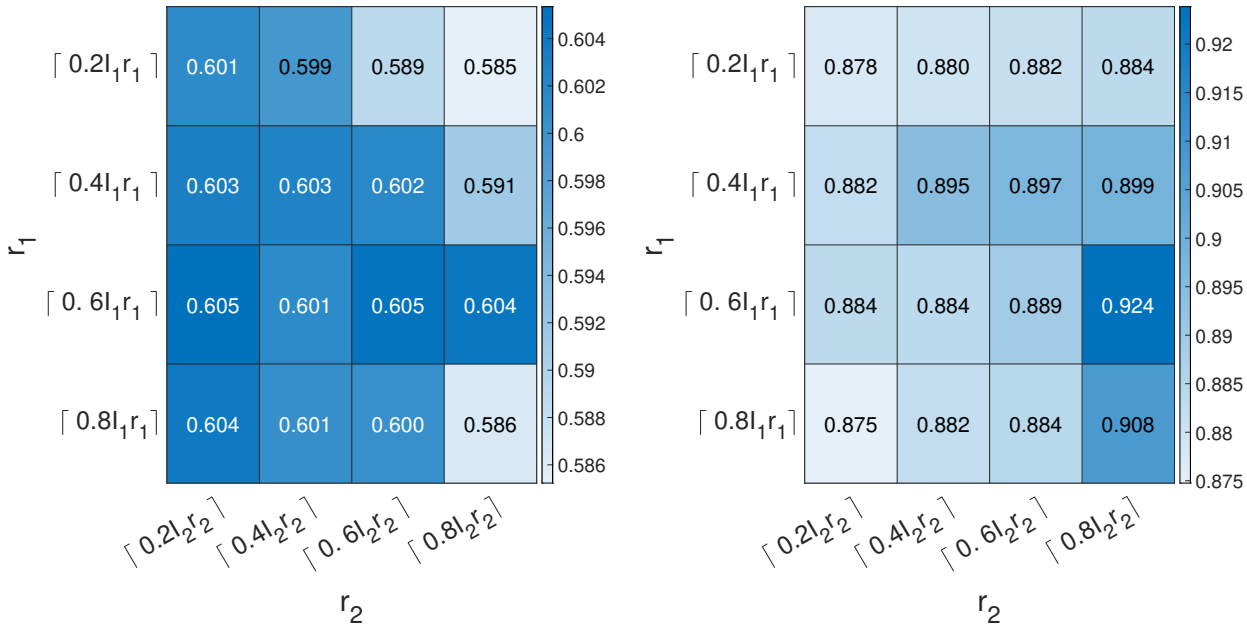
1) Fix $r_i=\frac{I_i}{2}$ $(\forall i\in[N-1])$ and $\alpha_q=1$ $(\forall q\in[Q])$, then alternately vary one of the sparsity level parameters λ_G, λ_i $(\forall i\in[N-1])$, and record the clustering performance under all possible parameter combinations, as shown in Figure 2(a).
2) Fix $\alpha_q=1$ $(\forall q\in[Q])$, $\lambda_G=1$ and $\lambda_i=1$ $(\forall i\in[N-1])$, then alternately vary one of the Tucker rank r_i $(\forall i\in[N-1])$, and record the clustering performance under all possible parameter combinations, as shown in Figure 2(b).

From Figure 2, we have the following observations.

1) For the sparsity level parameters, as λ_G decreases, the clustering performance of SMGNTD on the FashionMNIST dataset gradually improves and becomes less sensitive to both λ_1 and λ_2. Besides, when λ_G increases, the clustering performance of SMGNTD on the Pixraw10P dataset generally initially improves and then declines, while as λ_1 and λ_2 increase, the clustering performance of SMGNTD on the Pixraw10P dataset gradually becomes relatively stable.



(a) Clustering NMI with respect to the parameters λ_G and λ_i on the FashionMNIST (left) and Pixraw10P (right) datasets, respectively. The color hue of the point indicates the NMI value of the corresponding entry, and the mapping relationship between the NMI value and the color hue is displayed in the colorbar above.



(b) Clustering NMI with respect to the parameters r_i on the FashionMNIST (left) and Pixraw10P (right) datasets, respectively.

Figure 2: Parameter sensitivity analysis of SMGNTD on the FashionMNIST and Pixraw10P datasets.

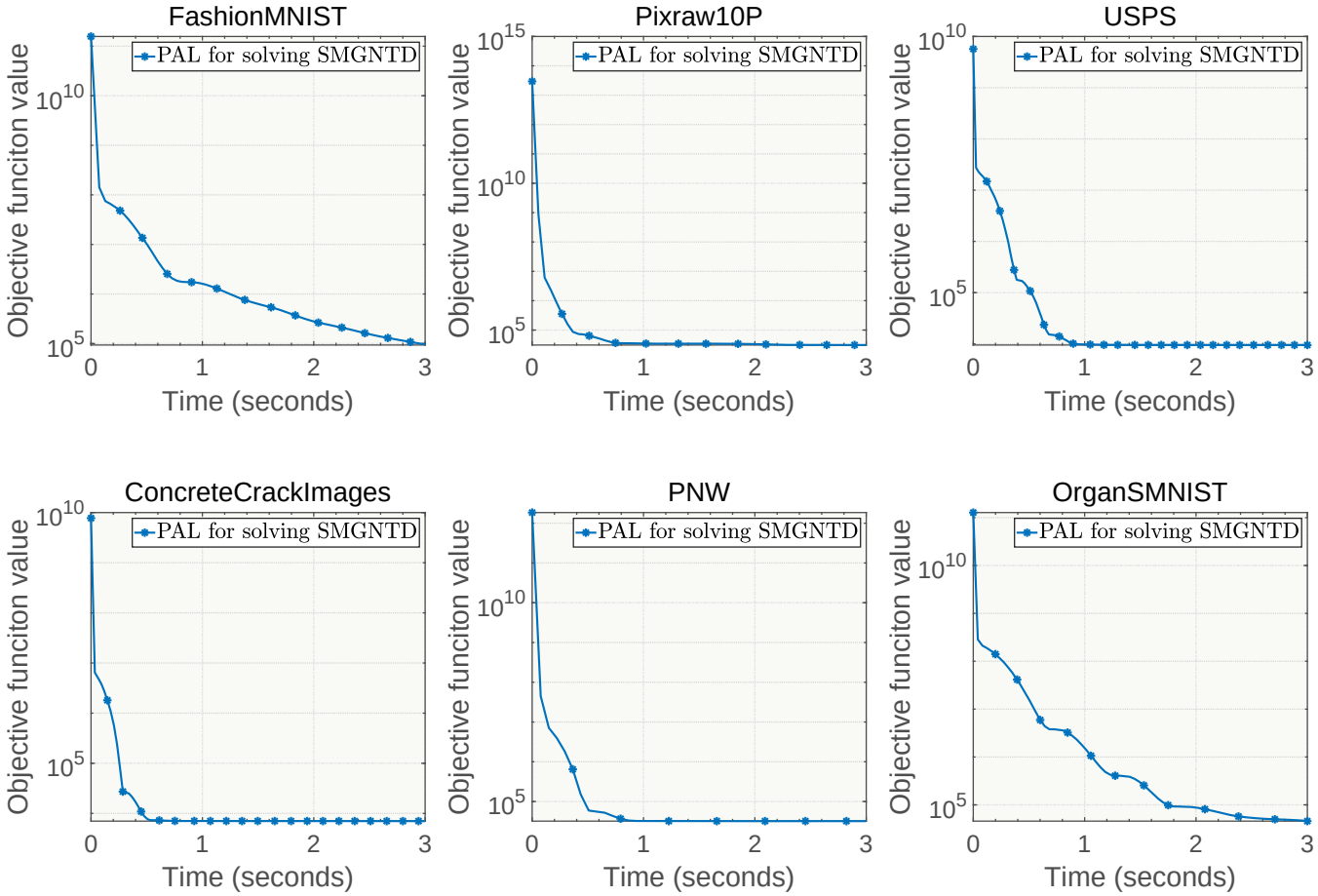


Figure 3: Convergence curves of the PAL algorithm for solving SMGNTD on six datasets.

2) For the Tucker rank, the clustering performance of SMGNTD on the FashionMNIST dataset initially improves and then declines as r_1 increases, while the clustering performance of SMGNTD on the FashionMNIST dataset shows a negative correlation with r_2. Besides, the clustering performance of SMGNTD on the Pixraw10P dataset also initially improves and then declines as r_1 increases, but the clustering performance of SMGNTD on the Pixraw10P dataset shows a positive correlation with r_2

C. Supplementary experiment 3: Convergence analysis

We also evaluate the convergence performance of our proposed PAL algorithm (Algorithm 1) for solving the SMGNTD. Figure 3 plots how the objective function value changes with respect to time for the PAL algorithm on several datasets. From Figure 3, we observe that the objective function value monotonically decreases to a stationary point, thereby validating the effectiveness and convergence of the PAL algorithm for solving the SMGNTD.

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